



United International University

School of Science and Engineering

Mid Term Examination; Year 2024; Trimester: Fall

Course: PHY 2105; Title: Physics; Sec: A-U

Full Marks: 30, Time: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled as per UIU disciplinary rules.
Questions no 1, 2, 3 are mandatory to answer. Answer anyone from question no 4 and 5.

1. (a) Why the oscillation of a mass spring system stops after some time? Graphically show the exponential decay of amplitude with time for the mass spring system. 2 CO1
- (b) The equation of displacement of a simple harmonic oscillator is $x = A \cos\left(\omega t + \frac{\pi}{2}\right)$. Plot velocity vs. angle and acceleration vs. angle graphically. What is the phase difference between the two graphs? 2 CO1
- (c) Robert found an inductor (L) and constructed a circuit with a charged capacitor (C) and a small resistor (R). Although he expected simple harmonic oscillation in the circuit, he found damped harmonic oscillation. Why he has observed this type of oscillation? What type of damping may be observed if he had connected a very high resistor? 2 CO1

2. (a) A particle executes SHM of amplitude 8 m when the particle is 6 m from its mean position, its acceleration is found to be 82 m/s^2 . 3 CO3
- (i) calculate the frequency of oscillation
- (ii) maximum velocity.

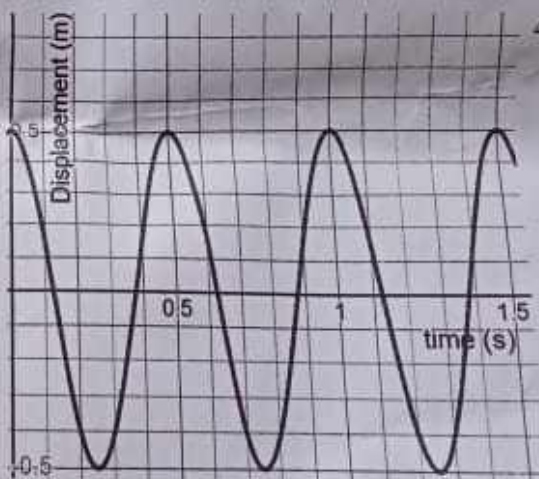
- (b) The adjacent graph shows the displacement of a mass spring system. 4 CO3

If the mass of the mass spring system is 40 g.

- (i) Calculate the displacement at $t = 20 \text{ s}$.

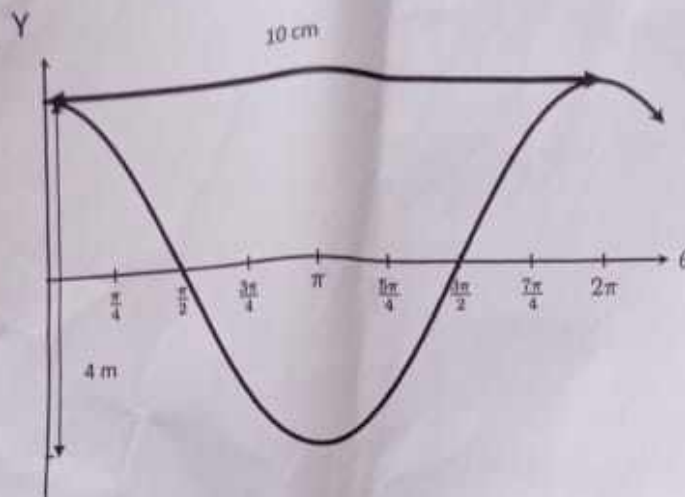
- (ii) The kinetic energy at $t = 4.5 \text{ s}$.

$$f = \frac{\omega}{2\pi}$$
$$\omega = 2\pi f, \quad \sqrt{\frac{k}{m}}$$
$$v = \omega \sqrt{A^2 - x^2}$$
$$\omega^2 = \sqrt{\omega^2 - \frac{x^2}{A^2}}$$
$$RLP = \frac{1}{\omega} < \frac{1}{\omega_0}$$
$$ove = \frac{R^2}{4L^2} > \frac{1}{\omega_0^2}$$
$$criti \quad \frac{R^2}{4L^2} = \frac{1}{\omega_0^2}$$
$$und on \quad \frac{R^2}{4L^2} < \frac{1}{\omega_0^2}$$



3. (a) For a damped oscillator, $m = 500 \text{ g}$, $k = 200 \text{ N/m}$, and $b = 69 \text{ g/s}$. The oscillator is released at $t = 0$ and the amplitude is 5 m. 3 CO3
- (i) Calculate the period of the motion.
- (ii) How long does it take for the amplitude of the damped oscillator to become 3 m?
- (b) Eduard constructed an RLC circuit with the value, $C = 8 \mu\text{F}$, $L = 0.5 \text{ H}$, respectively. 3 CO3
- (i) If we place the resistor with $2 \text{ K}\Omega$ what type of damping may be observed in that case?
- (ii) What will be the value of resistance R if he wants to produce critical damping.

(c) A wave traveling in the positive x direction has a frequency of $f = 20$ Hz as shown in the figure below. Calculate the (i) wavelength, (ii) wave speed (iii) equation of displacement of the particle of medium. 3 CO3



4. (a) A mass spring system with spring constant k is pressed up to its amplitude A and released at $t = 0$. If the system is oscillating in simple harmonic oscillation, derive differential equation for the mass spring oscillator and find out the possible solution of displacement for the system. 4 CO2

(b) For a mass-spring system oscillating in SHM, the equation of displacement is,

$$y = A \cos(\omega t + \frac{\pi}{4})$$

4 CO2

Show that potential energy and kinetic energy of the system are time dependent. Drive the necessary equations and plot a graph of energy vs. time to validate your answer.

5. (a) (i) Derive the equation of a progressive wave. 4 CO2

(ii) Show that, $\frac{dy}{dt} = -v \frac{dy}{dx}$.

(b) (i) Derive the differential equation of damped harmonic oscillation (DHM) for a mass spring system. 4 CO2

(ii) Find out the conditions of different types of damping and represent them graphically. [Draw amplitude vs. time graph for each type]

CO1: Define different physical quantities with examples CO2: Derive/Show the various equations of SHM, DHM, wave motion etc.
CO3: Evaluate different numerical problems based on the basic characteristics of SHM, DHM.